

A 7-nm Four-Core Mixed-Precision AI Chip With 26.2-TFLOPS Hybrid-FP8 Training, 104.9-TOPS INT4 Inference, and Workload-Aware Throttling

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Abstract—Reduced precision computation is a key enabling factor for energy-efficient acceleration of deep learning (DL) applications. This article presents a 7-nm four-core mixed-precision artificial intelligence (AI) chip that supports four compute precisions—FP16, Hybrid-FP8 (HFP8), INT4, and INT2—to support diverse application demands for training and inference. The chip leverages cutting-edge algorithmic advances to demonstrate leading-edge power efficiency for 8-bit floating-point (FP8) training and INT4 inference without model

accuracy degradation. A new HFP8 format combined with separation of the floating- and fixed-point pipelines and aggressive circuit/architecture optimization enables performance improvements while maintaining high compute utilization. A high-bandwidth ring protocol enables efficient data communication, while power management using workload-aware clock throttling maximizes performance within a given power budget. The AI chip demonstrates 3.58-TFLOPS/W peak energy efficiency and 26.2-TFLOPS peak performance for HFP8 iso-accuracy training, and 16.9-TOPS/W peak energy efficiency and 104.9-TOPS peak performance for INT4 iso-accuracy inference.

Index Terms—Approximate computing, artificial intelligence (AI), deep neural networks (DNNs), hardware accelerators, machine learning (ML), reduced precision computation.

I. INTRODUCTION

ARTIFICIAL intelligence (AI) techniques have shown explosive progress over the past decade across a wide spectrum of applications, including computer vision, natural language processing, and object detection—even surpassing humans on many tasks. As such, AI workloads have become prominent across the full spectrum of computing platforms from mobile and IoT edge devices to data centers. To meet the ever-growing computational demands of state-of-the-art AI algorithms, research focus has shifted to dedicated hardware accelerators that provide orders of magnitude higher throughput and energy efficiency than traditional programmable hardware such as CPUs. As a result, tremendous progress has been made in recent years to push the boundaries of compute efficiency and throughput as demonstrated by published works from academia [1]–[5] and industry [6]–[9] alike.

To continue performance scaling to fuel future breakthroughs in AI applications, a key area of focus has been bit precision scaling by exploiting the inherent error resiliency of AI algorithms to enable reduced precision computation for both training and inference [10]–[15]. Reduced precision

Manuscript received May 14, 2021; revised August 8, 2021 and September 27, 2021; accepted October 1, 2021. Date of publication November 10, 2021; date of current version December 29, 2021. This article was approved by Associate Editor Jun Deguchi. (*Corresponding author: Sae Kyu Lee.*)

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Digital Object Identifier 10.1109/JSSC.2021.3120113

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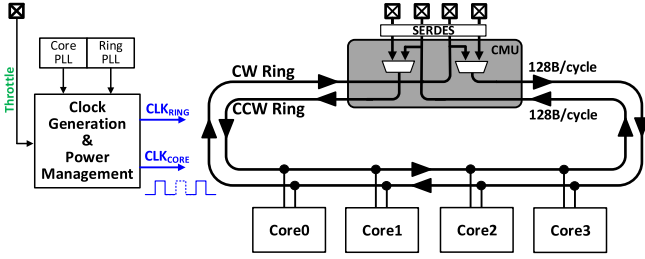


Fig. 1. Four-core mixed-precision AI chip architecture.

offers several key advantages for hardware acceleration. It has a well-defined impact on hardware efficiency by scaling compute density while reducing memory footprint and bandwidth requirements commensurate to the bit precision. It also preserves the regularity of compute that mitigates unnecessary control overhead and complexity [3].

However, precision scaling also poses several key challenges to harnessing its full benefits, most important of which is the need to maintain model accuracy equivalent to high-precision computation, which is critical for wide deployment. In addition, it poses a burden on the overall system design to improve bandwidth, architecture, and power management to feed and support more compute engines to achieve high sustained utilization and performance within the given power envelope.

This work presents a mixed-precision four-core AI chip, shown in Fig. 1, which addresses these challenges and demonstrates leading-edge efficiency for low-precision training and inference without model accuracy degradation [16]. The AI chip leverages recent algorithmic advances to introduce a new Hybrid-FP8 (HFP8) format in addition to the FP16 format for training and inference and supports INT4 and INT2 formats for highly scaled inference. In addition, the chip includes a high-bandwidth ring interconnect for efficient many-core scalability and employs workload-aware power control to maximize the end-to-end performance.

The remainder of this article is organized as follows. Section II gives an overview of the algorithmic advances in precision scaling for both training and inference leveraged in this work. Section III provides an overview of the AI chip architecture and its main features. Section IV describes the AI chip implementation, while Section V presents the measurement results from the 7-nm AI chip. Section VI concludes this article.

II. PRECISION SCALING FOR TRAINING AND INFERENCE

This section describes the algorithmic advances that are leveraged in this work, which enables iso-accuracy model training in the 8-bit floating-point (FP8) and inference in the 4-bit fixed-point (INT4) format.

A. Iso-Accurate HFP8 Training

While FP16 training has enjoyed wide adoption [17]–[19], scaling precision below FP16 has proven challenging due to inherent precision requirement differences in the forward and

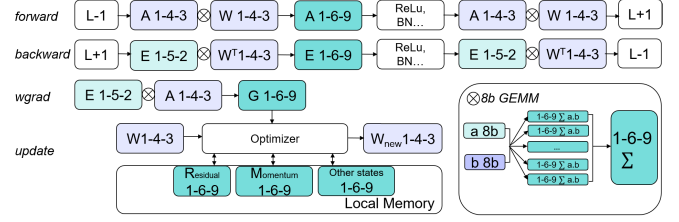


Fig. 2. HFP8 formats used in forward, backward, gradient computation, and weight update phases of training in a ResNet layer [13].

backward training phases and the need to maintain gradient fidelity [20]. Recently, an HFP8 format was introduced to overcome this limitation and achieve iso-accuracy model training in the FP8 format compared to an FP32 baseline [13]. The HFP8 format uses two different FP8 (sign, exponent, and mantissa) representations, FP8(1, 4, 3) and FP8(1, 5, 2), to cater to the differing dynamic range and precision requirements of the forward and backward passes of deep neural network (DNN) training. Weights and activations use the FP8(1, 4, 3) format, which provides higher precision using the extra mantissa bit, while errors in backward passes are represented using the FP8(1, 5, 2) format to provide a higher dynamic range. In addition, FP8(1, 4, 3) uses a programmable exponent bias, which enables DNN layers to be represented by different dynamic ranges within the same (1, 4, 3) format constraint. Fig. 2 shows the formats used for computation in a forward, backward, and update pass of a backpropagation-based training of a ResNet [21] layer. In the forward pass, both operands [weights (W) and activations (A)] use the FP8(1, 4, 3) format, while in backward and update passes, one operand [error (E)] uses FP8(1, 5, 2). It should be noted that certain computations, such as activation functions and optimizer algorithms for weight update, are sensitive to precision and still require FP16 computation, necessitating the need to maintain FP16 support. As such, 8-bit generic matrix multiplication (GEMM), shown in the inset, produces 16-bit results for downstream processing, while parameters, such as momentum used for weight update (W_{new}), are represented in FP16. HFP8 achieves iso-accuracy training compared to an FP32 baseline across a wide range of workloads spanning image classification, natural language processing and translation, speech, and object detection [13].

B. Iso-Accurate INT4 Inference

Recently, iso-accurate INT4 inference has been demonstrated using two quantization techniques for activations and weights: Parameterized Clipping acTivation (PACT) [14] and statistics-aware weight binning (SAWB) [15]. For activation quantization, PACT finds the optimal quantization scale by introducing a parameterized clipping level that balances quantization error and clipping error. The clipping level is learned via backpropagation to find the optimal clipping level independently for each DNN layer. For weights, SAWB exploits the statistical characteristics of weight distribution (the first and second moments) to determine the quantization scale to better capture the dispersion of weights. These techniques

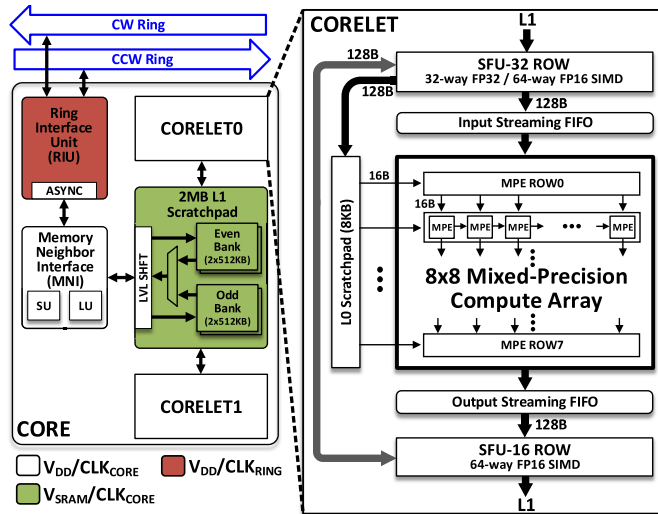


Fig. 3. Mixed-precision AI core architecture block diagram.

together enable both weights and activations to be quantized to 4 bits while achieving negligible accuracy degradation compared to full-precision network representations. These techniques have also been applied to scale down to INT2 inference albeit with some accuracy loss (1%–2%) [15].

III. FOUR-CORE MIXED-PRECISION AI CHIP

Fig. 1 shows the block diagram of the 7-nm four-core mixed-precision AI chip that exploits the precision scaling techniques discussed in Section II to provide support for robust iso-accurate training in FP16 and HFP8 formats, as well as highly scaled inference in the INT4 and INT2 formats [16]. The chip consists of four mixed-precision AI cores connected to high-bandwidth bi-directional clockwise (CW) and counterclockwise (CCW) rings, each with 128-B/cycle bandwidth. A chip management unit (CMU) can either configure the rings to be closed within a single-chip or connect multiple chips to form a larger ring to scale up to large-scale many-core systems. The cores and the bi-directional rings operate in independent clock domains using dedicated core and ring phase-locked loops (PLLs), with asynchronous clock domain crossings handled by the cores at the ring interface. This allows the rings to be tuned independently of the cores to optimally balance power and performance for compute and data movement. Finally, the chip also includes an on-chip power control module for workload-aware throttling to fully utilize the chip power budget to maximize the application performance.

A. Core Architecture

Fig. 3 presents the block diagram of the mixed-precision AI core. The core is a third-generation design [16], [22], [23] composed of two corelets with a shared software-managed 2-MB L1 scratchpad memory. Each corelet contains an 8×8 2-D systolic array of mixed-precision engines (MPEs) optimized to accelerate convolution and matrix multiplication operations critical in AI workloads. MPE offers low-precision iso-accurate training and inference of deep learning (DL)

models while supporting four precision modes to meet diverse application demands. The MPEs support the HFP8 and FP16 (DLFloat16 [17]) formats for both training and inference, while INT4 provides highly scaled iso-accurate inference. The MPE also supports INT2 inference for applications in which some loss of accuracy is permissible. The MPE array implements a systolic dataflow with data from an 8-kB L0 scratchpad feeding data from the west, while north and south data are accessed by input and output streaming first in first out (FIFO) buffers.

Each corelet includes two rows of special function units (SFU-16/SFU-32) to provide dedicated support for auxiliary vector operations, such as activation functions, pooling, and gradient reduction to provide full end-to-end application coverage in the core. Both SFU rows support 64-way single-instruction multiple-data (SIMD) FP16 computations, while SFU-32 can also perform 32-way SIMD FP32 computations.

To maximize data reuse, the core uses a two-scratchpad hierarchy: an 8-kB L0 scratchpad in each corelet that feeds data to the MPE array from the west and a shared 2-MB L1 scratchpad organized as even and odd banks. The large capacity L1 scratchpad is shared between two corelets for more efficient use of scratchpad capacity and to better exploit the data-sharing behavior of AI workloads. The L1 and L0 scratchpad sizes were chosen to maximize data reuse while balancing the on- and off-chip bandwidth constraints with scratchpad area and power. Since the external DRAM bandwidth is typically an order of magnitude lower than that of on-chip, the 2-MB L1 scratchpad capacity was designed to provide sufficient aggregate on-chip storage to enable large activation data structures of many layers to fit in the distributed L1 scratchpads and allow intermediate outputs to stay on-chip to avoid costly off-chip DRAM accesses while balancing the capacity requirements against per-core area and power constraints. In contrast, L0 benefits from high-bandwidth on-chip access to L1 (128-B/cycle), which enables efficient data reuse with 8-kB L0 scratchpad in each corelet. The memory neighbor interface (MNI) handles data transfers between the L1 scratchpad and external memory or other cores via the ring interface unit (RIU) through the ring bus. The core implements a decentralized architecture that decouples compute and dataflow that enables the data fetch latency to be effectively hidden by double buffering data in the L1 scratchpad overlapped in time with computation.

B. MPE Overview

The performance and efficiency of the 2-D systolic mixed-precision compute array shown in Fig. 3 are critical to the overall performance of the AI core. The MPE row, shown in Fig. 4(a), consists of eight MPEs with a shared instruction fetch (IF) stage for the entire row. Since typical dataflows do not require the diagonal flow of operands, all MPEs within a row support systolic execution of the same instruction sequence, where decoded instructions from the shared IF stage are propagated along with data in each cycle. This eliminates the need for separate instruction buffers in each MPE to minimize the control logic overhead, shortens wire lengths,

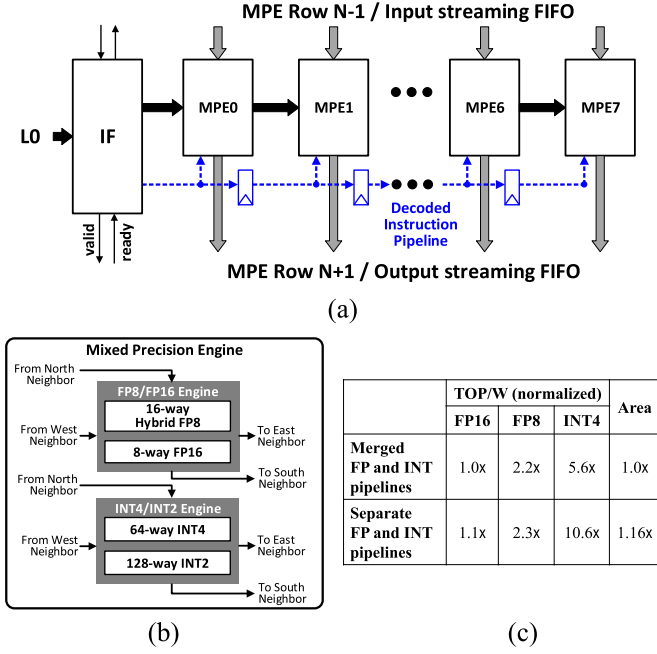


Fig. 4. (a) Simplified block diagram of the MPE row illustrating the dataflow and instruction pipeline. (b) MPE block diagram with separate FP and FX pipelines. (c) Summary of area-efficiency tradeoff of merged versus separate pipelines.

and enables the instruction issue (IS) stage to complete in a single cycle resulting in significant gains in compute and area efficiency. In addition, the MPE instruction set architecture (ISA) is carefully limited to only the instructions that support matrix multiplication and convolution to further improve compute efficiency.

Supporting multiple compute precisions in the MPE, shown in Fig. 4(b), presents a unique challenge due to differing requirements of each format for optimal performance. To better cater to the needs of each precision, the floating-point (FP) and fixed-point (FX) pipelines were decoupled to implement separate compute engines. As summarized in Fig. 4(c), a unified compute pipeline, while more area-efficient, results in limited energy efficiency due to the complexities associated with supporting multiple compute precisions within a single datapath. Separate pipelines enable each unit to be optimized independently and tailored to its target application. This results in overall improvement in energy efficiency, especially for integer computation with $2\times$ improvement in INT4 efficiency while incurring a 16% area overhead.

C. FP Engine

Fig. 5 shows the simplified block diagram of the FP engine containing eight-way SIMD FP units (FPUs) that perform FP16 and HFP8 computations with 128-bit datapath. The FP engine implements a three-stage execution pipeline with carefully curated datapath flexibility in the operand muxes to support efficient mapping of a comprehensive range of dataflows in the MPE array. Fig. 5(b) summarizes the datapath flexibility in the operand muxes and presents the examples of different dataflow mappings for each of the forward, backward,

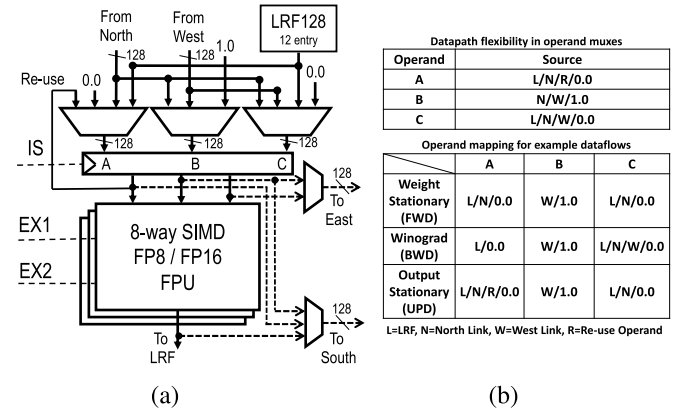


Fig. 5. (a) Block diagram of the FP engine with eight-way SIMD FPUs. (b) Summary of datapath flexibility for efficient dataflow mappings.

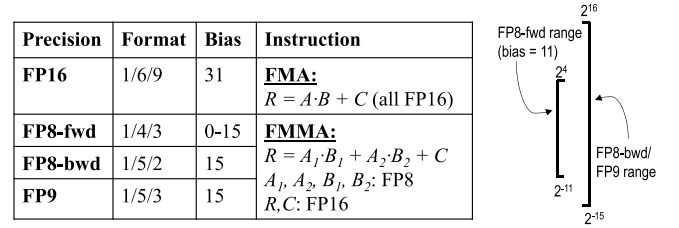


Fig. 6. FP formats and MAC instruction supported in FP engine (left) and illustration of configurable bias for FP8-fwd format to enable programmable adjustment of the dynamic range within the 8-bit limit, shown for example FP8-fwd bias of 11 (right).

and update passes of the backpropagation algorithm as an illustration. The datapath flexibility enables the core to map a variety of dataflows to cater to the workload requirements needed to achieve high sustained compute utilization in the MPE array.

As discussed in Section II, the HFP8 format uses tensors of two different FP8 formats (sign, exponent, and mantissa) for iso-accurate training: the (1, 4, 3) format (FP8-fwd) for activations and weights and the (1, 5, 2) format (FP8-bwd) for errors during backward and update phases of training. Fig. 6 summarizes the precision formats and multiply-accumulate (MAC) instructions supported in the FP engine. The FP8-fwd format has a programmable bias between 0 and 15, which enables programmable adjustment of the FP8-fwd dynamic range. In the FP8 mode, the FP engine partitions the 16-bit SIMD lanes in two to perform a fused-multiply-multiply-accumulate (FMMA) instruction with A_1, A_2, B_1, B_2 in one of two FP8 formats, while the addend (C) and result (R) are in the FP16 format. With each instruction performing two multiplications and two additions, the FP8 mode doubles the throughput and effective bandwidth compared to the FP16 mode while preserving the interface datapath width. To minimize the hardware overhead of supporting two different FP8 formats, the FPU employs a unified FP9 format of (1, 5, 3) with input operands in either (1, 4, 3) or (1, 5, 2) formats converted on-the-fly to the FP9 format at the boundary of the MPE array. This enables the weights and activations to be stored in the L0 and L1 scratchpads in the FP8 format with no additional burden on scratchpad capacity.

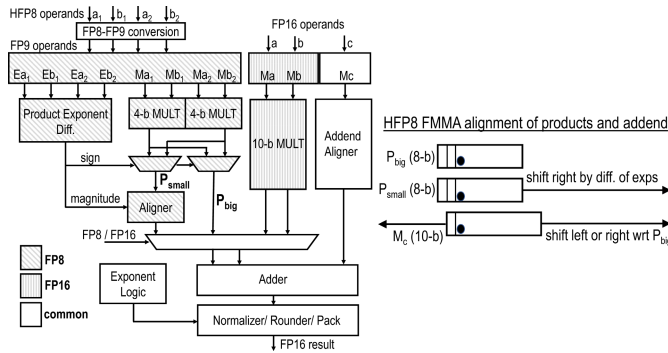


Fig. 7. Block diagram of the FPU pipeline showing datapath for FP16 and HFP8 computation (left) with illustration of product term alignment for FMMA computation in HFP8 mode (right).

Fig. 7 presents the FPU datapath highlighting the multiplication and alignment logic for HFP8 computation, FP16 multiplier path, and addend aligner common to both formats. Since both the HFP8 and FP16 datapaths produce FP16 results, the compute paths merge at the adder to share the downstream logic. A key challenge of implementing the FMMA instruction compared to the FMA instruction is that it requires three mantissa terms to be aligned with respect to each other for accurate intermediate sum calculation. While this can be achieved by aligning the three terms to an absolute reference, it requires large aligners with wide adders and shifters that incur large area and power penalty. To overcome this limitation, the FPU aligns the smaller product term (P_{small}) and the addend (C) to the exponent of the larger product term (P_{big}), which requires the products of the two 4-bit multipliers to be first compared and sorted before alignment. Thus, the exponent difference of the product terms is also calculated to shift and align the smaller product, as shown at the output of the 4-bit multipliers in Fig. 7. Since the HFP8 datapath only requires 4-bit multipliers, the HFP8 multiplicand path has comparable logic depth to FP16 even with the additional complexity at the multiplier output. In parallel to the exponent difference calculation, we also pre-compute two addend mantissa shift amounts and then choose the one corresponding to P_{big} as the control input to the addend mantissa shifter, thus only adding additional logic complexity of only one MUX stage in the aligned addend generation path. It should be noted that all accumulation operations in the FPU are performed exclusively using the FP16 format, enabled by hierarchical chunk-based accumulation [24]. By accumulating a large number of FP numbers hierarchically in groups using FP16, convergence quality as could otherwise be achieved through serial FP32 accumulation can be preserved, thereby eliminating the need for expensive FP32 accumulation.

Finally, to maximize energy efficiency in the FPU, all unused datapaths in the FPU are meticulously clock- and data-gated to minimize unnecessary power consumption. In addition, the FPU provides support to exploit the abundant sparsity in AI workloads by skipping the entire FPU pipeline when multiplicands are zero and simply forward the addend to the result. Overall, the HFP8 computation consumes similar power to that of FP16, resulting in a $2\times$ TFLOPS/W improvement.

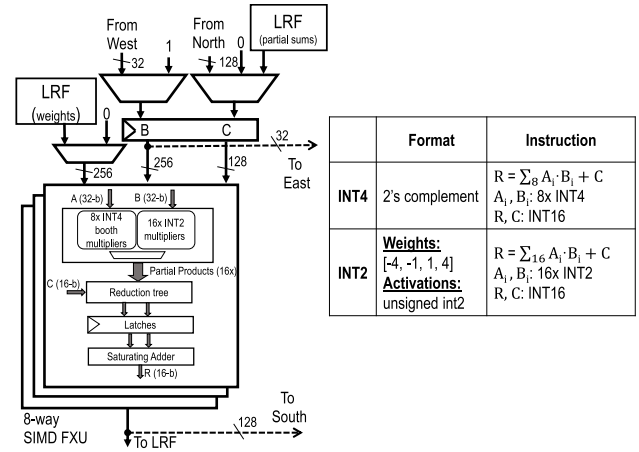


Fig. 8. Block diagram of the FX engine with supported data formats and MAC instructions.

D. FX Engine

Fig. 8 presents the FX engine datapath, which consists of eight-way SIMD fixed-point units (FXUs) that perform either INT4 or INT2 MAC operations with INT16 addends and results. As previously mentioned, decoupling the FX and FP engines allowed a ground-up design optimization of the FX engine to enable a compact datapath, shorter execution latency (two-cycle execution versus three-cycle execution for the FP engine), shorter wires, and reduced latch stages. In addition, since the FX engine is specifically designed to target only DNN inference, operand MUXing and latch stages were further simplified and tailored for inference-only dataflows to provide further area and power savings compared to the FP engine that supports backward and update passes.

As a result, the INT4 operation is $4\times$ more energy efficient than FP8 operations, as shown in Fig. 4(c). This provided an opportunity to double the integer datapath by instantiating $2\times$ more integer compute units within the same power budget to achieve $2\times$ TOPS than traditional sub-word SIMD, resulting in an eight-way INT4 and 16-way INT2 MAC datapath within each FXU. Since this doubles the internal FX engine datapath width to 256 bits, provisioning sufficient bandwidth to feed the double-wide datapath becomes critical. To balance bandwidth constraints with the increase in compute, the FX engine architecture is augmented to allow two local register file (LRF) accesses with a single MAC instruction, while dataflow mapping is enhanced to reuse the 32-bit operand from the west link across the entire eight-way SIMD FXUs. Although the FXU internal datapath has doubled to 256 bits, FXU still produces 16-bit results to maintain the 128-bit interface between the MPEs.

As discussed in Section II, INT2 inference incurs a slight accuracy degradation compared to the INT4 mode. However, for applications where some accuracy loss is permissible, INT2 inference may provide an acceptable tradeoff by providing $2\times$ the throughput and energy efficiency of INT4 mode.

E. Special Function Unit

Fig. 9 presents the block diagram of the SFU row, consisting of eight SFUs with central IF and IS units. The SFUs support

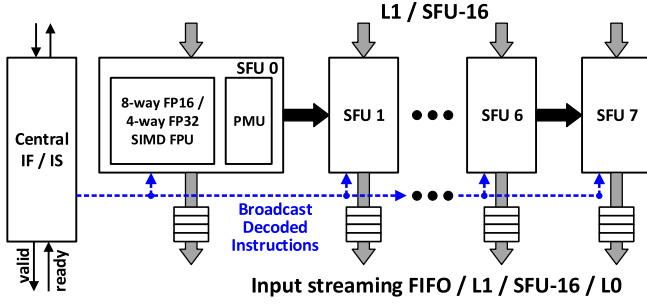


Fig. 9. Simplified block diagram of the SFU row (SFU-32).

a comprehensive range of non-linear activation functions such as ReLU, leaky ReLU, sigmoid, and tanh in both fast and accurate versions, as well as pooling, normalization, interslice reduction operations, data manipulation, and hierarchical chunk-based accumulation for numerical stability [24]. Each SFU includes eight-way SIMD FP16 FPU for 64-way SIMD FP16 computations across the entire SFU row and a permute unit (PMU) for common 128-bit operations such as merge. SFU-32 row can also perform FP32 computations by reconfiguring two 16-bit FPUs into a 32-bit FPU pipeline for computations that require higher bit precision. The SFUs also perform the conversion between the IEEE FP16 and BFloat16 formats and the DLFloat16 format used in the FPUs. In contrast to the systolic dataflow in the MPE row, SFU rows support SIMD compute mode across the entire row, which requires centralized IF and IS units to broadcast decoded instructions to all the SFUs in the row. This necessitates a deeper execution pipeline with a two-cycle IS stage.

As throughput in the MPE array scales with reduced precision computations, the high-precision vector operations start to occupy a larger percentage of the overall runtime. Therefore, each corelet includes two SFU rows to better balance the runtime spent in high-precision vector operations versus low-precision convolution and matrix multiplication.

F. Workload Mapping

For efficiency, we utilize an in-house end-to-end software stack to automatically map the AI workloads onto the chip [25]. The software stack includes a graph compiler that performs a systematic design space exploration to identify the best workload mapping to the cores on the AI chip. The chip itself is flexible and imposes no constraints on how the layers are mapped onto the cores, and the compiler modulates the dimension that is split across the cores based on the computational characteristics of a given layer.

As discussed in Section III-C, the MPE array has the datapath flexibility to support a wide spectrum of dataflow mappings on the array. Fig. 10 shows workload mapping for weight- and output-stationary dataflows commonly used to map a DNN layer on the MPE array. Our weight-stationary flow as shown in Fig. 10(a) maps the input channel (C_i) dimension along the rows and output channel (C_o) dimension along the column and SIMD dimensions. We avoid mapping the feature map ($H \times W$) or kernel map ($K_i \times K_j$) dimensions

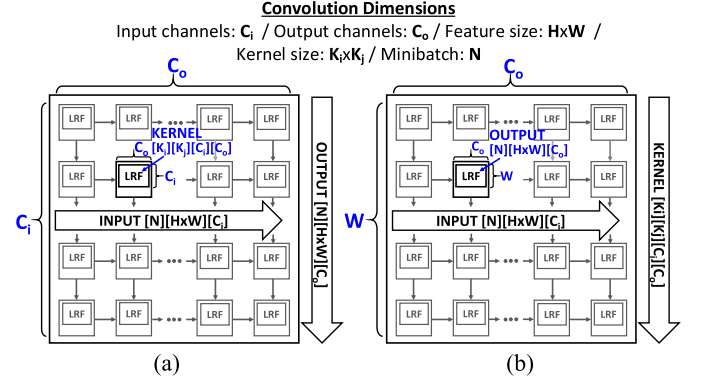


Fig. 10. MPE array workload mapping for (a) weight-stationary dataflow and (b) output-stationary dataflow.

spatially as they are typically not multiples of the hardware dimensions and result in row/column under-utilization. The inputs are streamed from west to east and are broadcast to each column and SIMD through the systolic array. The outputs are streamed from north to south. The weights are stored in the LRF. Each MAC unit takes input from the west, multiplies it with the weight stored in its LRF, and updates the output coming from the north and sends the result to the south.

For layers with very few input channels (e.g., first layer), output-stationary flow can be used, as shown in Fig. 10(b). This maps the feature map column dimension (W) along the rows of the MPE array, with each row computing a different column pixel of the output. The columns and SIMD lanes still compute different output channels (C_o) with inputs streamed from west to east, but weights are now streamed from north to south. Each MPE accumulates the results in the LRF and stores it back to the L1 scratchpad after all partial products are accumulated.

G. High-Bandwidth Ring Interconnect

The chip implements a high-bandwidth bi-directional ring interconnect for efficient data communication to support scaling to a large system using multiple AI chips. The RIU in conjunction with the MNI in each core, composed of separate programmable load and store units (MNI-LU and MNI-SU), facilitates data transfers between the L1 scratchpad and the external memory and other cores, as shown in Fig. 11(a). Each entity (cores/memory) on the ring is associated with a software-assigned unique identification tag for communication. Every data transfer is initiated by load/store requests from MNI-LU/SU, with all packets (request and data packets) containing the source and destination information. For unicast communication, the packet simply contains the destination ID and the RIU is configured to inject the packet in the ring that has the least number of hops to the destination. The software-assigned ID tags allow the requests to be tracked to enable multiple outstanding requests on the ring and out-of-order data returns.

The MNI-LU and SU also support multicast communication to memory and other cores to further exploit the high degree of parallelism and data sharing behavior of AI workloads while

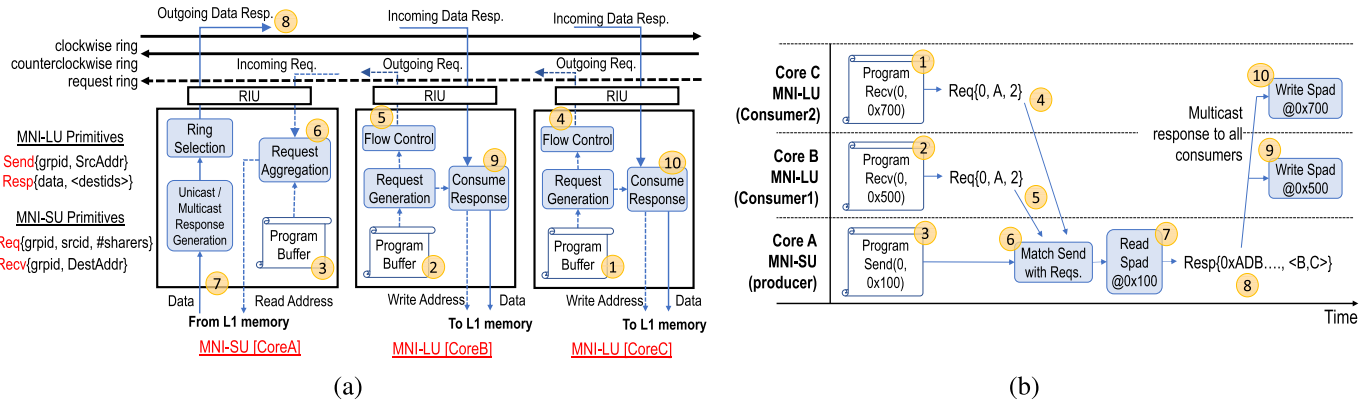


Fig. 11. Example multicast transfer between consumer cores B and C receiving data from producer core A illustrating (a) flow sequence and (b) corresponding timing diagram.

alleviating congestion on the ring. For multicast communication, the transfer packet contains a bit vector constructed to indicate the position of the destinations within the ring. The RIU of the producer core is configured to randomly inject the packet in either of the two rings with equal probability such that their load is loosely balanced, while each RIU on the ring compares the destination information with its ID to know whether it is a recipient and the RIU of the last destination removes the packet from the ring. Fig. 11 shows an example illustrating multicast data transfer from core A to cores B and C, presenting each step and the associated timing diagram. Multicast data transfer involves consumer cores (cores B and C) making individual *recv* request to the producer core (core A) with a common identification tag and the number of consumers, while core A executes a corresponding *send* primitive with matching identification tags (steps 1–3). Next, core A aggregates all requests before reading the L1 scratchpad and sending the data onto the ring using the same identification tag along with the list of consumers (steps 4–8) to be consumed by the requestors (steps 9 and 10).

To fully utilize the bi-directional CW and CCW rings, the MNI-LU is designed to receive up to two data returns at any time to provide overall high bandwidth for data returns to the core. To avoid bank collisions in the L1 scratchpad, requestors inform whether the data are for even or odd L1 bank, and the RIU employs a ring arbitration scheme for slot reservation that ensures two concurrent data returns in the CW and CCW rings always target opposite banks. Furthermore, the ring enables multiple independent segments to be used concurrently, enhancing the effective intra-chip bandwidth.

Finally, to optimally balance the power/performance of compute with communication requirements, the bi-directional rings operate on an independent clock domain with RIU providing support for asynchronous clock domain crossings.

H. Workload-Aware Power Management via Clock Throttling

AI workloads can exhibit large variation in power consumption and execution time due to heterogeneity in compute characteristics across layers, with the number of input and output channels and the dimensions of each feature

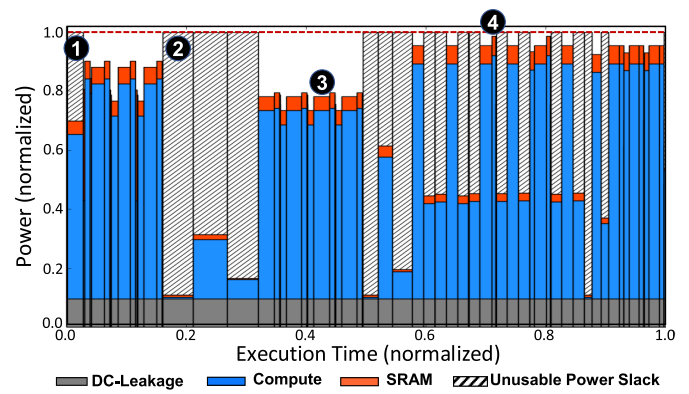


Fig. 12. Layer-by-layer power breakdown of ResNet-50 inference plotted against layer execution time to demonstrate heterogeneity in power consumption and execution time. Four layer types are highlighted to illustrate differences in layer characteristics. 1: memory-bound Conv. 2: memory-bound ReLU. 3: conv layer with layer dimensions limiting data reuse. 4: conv layer with no DRAM data dependency or kernel reloading.

being vastly different [26]. Moreover, AI workloads are unique in that these characteristics are known in advance at compilation time. In this work, we exploit this characteristic to implement a workload-aware software/hardware co-optimization methodology that combines software-controlled operating voltage–frequency selection with an on-chip frequency throttling mechanism to maximize the performance within the given power envelope. To demonstrate the layer-specific power variation of an end-to-end network, Fig. 12 shows the layer-by-layer power breakdown of the AI cores running the ResNet-50 benchmark plotted against the execution time for a projected system of 32 AI-cores attached to an external DRAM memory with 192-GB/s bandwidth. The system is modeled using chip measurements in conjunction with our in-house architectural simulator tool [25]. Typically, the average power consumed by a layer over its execution time is dictated by the following factors.

- 1) The layer is memory bound and core stalls waiting for data from DRAM, a common case if the previous layer's output cannot be contained in the distributed L1 scratchpad, or for the first convolution layer of a network (layers 1 and 2).

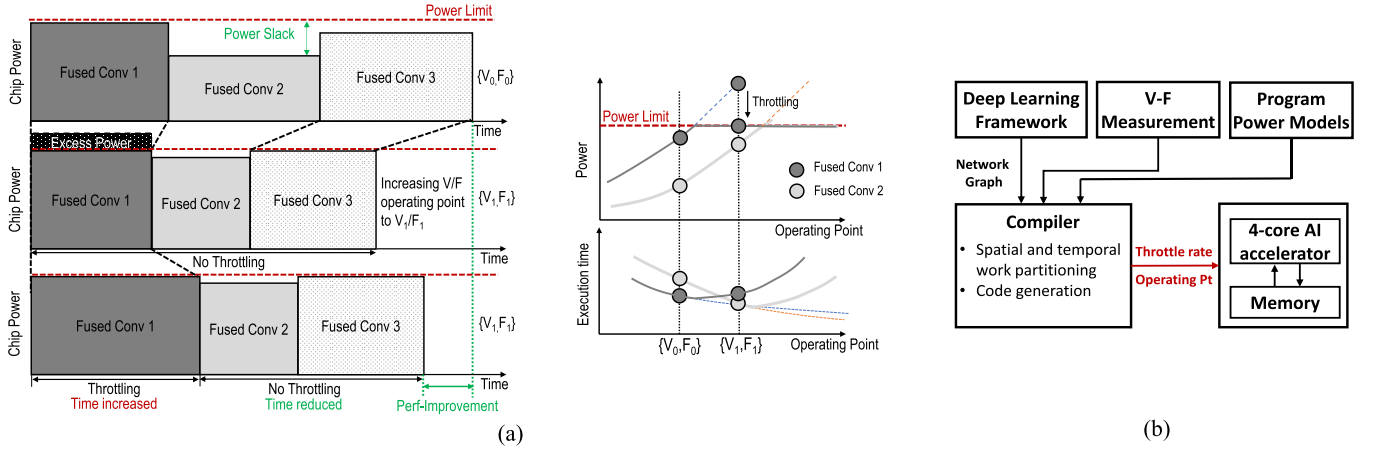


Fig. 13. (a) Illustration of end-to-end performance improvement via workload-aware operating point selection and frequency throttling. (b) Block diagram of the compiler-driven operating point and throttling rate selection methodology.

- 2) The dimensions of a convolution layer limit the data reuse requiring kernel reloading. For example, a smaller input feature map size reduces the reuse of the kernel in a weight-stationary dataflow (used in the AI core) (layer 3).

Convolution layers that have no DRAM data dependency or kernel reloading requirements (such as layer 4) will exhibit the highest utilization and power consumption and run close to the system power limit. However, power consumption for many layers will fall below the power limit due to these factors, thus creating a power slack. While power slack of the memory-bound layers (layers 1 and 2) cannot be used to speed up the layer, power slack due to layer dimensions (layer 3) can be exploited to improve execution time.

Fig. 13(a) shows an illustration of the workload-aware power management methodology for an example network with three synthetic fused conv layers that are not limited by kernel re-loading nor memory-bound. Rather than running the cores at V_0/F_0 dictated by Fused Conv 1 (top), we can exploit the power slack in layers 2 and 3 to instead operate the chip at higher voltage/frequency (V_1/F_1) (middle). While this pushes layers 2 and 3 close to the power limit to speed up execution, without throttling, it will also push layer 1 to operate over the power limit. Therefore, clock frequency is throttled for layer 1 to stay within power limit (bottom). Despite the increase in the execution time of layer 1 due to throttling, the increase in operating voltage and frequency leads to overall improvement in end-to-end application performance due to the speedup of layers 2 and 3.

A key advantage of this scheme is that the optimal operating point as well as throttling rate for each layer can be generated offline, as workload and individual layer details are known at compilation time. During silicon characterization, chip power and frequency can be measured as a function of voltage and utilization to develop a detailed power/performance model. As presented in Fig. 13(b), the compiler can use this characterization to determine the optimal operating voltage–frequency for the workload as well as throttle rates for each layer that maximize end-to-end application performance—an analysis

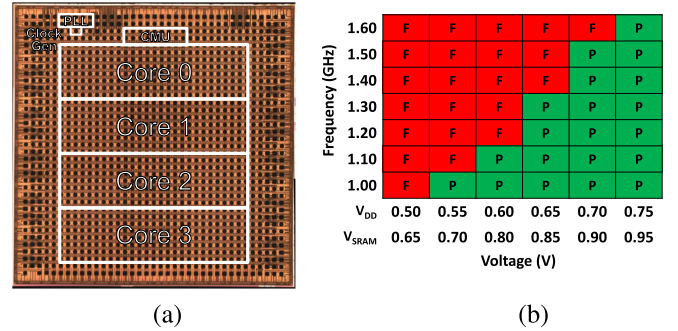


Fig. 14. (a) Annotated die photograph. (b) Shmoo plot.

that can be performed offline. Clock frequency throttling provides a simple and low-cost yet effective method for modulating power and performance within this framework since power/performance effects based on clock throttling rate are predictable. In this work, clock frequency throttling is implemented by using a 5-bit external throttle rate signal to skip clock edges within a 32-cycle window, producing an averaged clock frequency throttling effect.

IV. AI CHIP IMPLEMENTATION

The 36-mm² four-core mixed-precision AI chip [16] was fabricated in 7-nm technology. Fig. 14(a) presents the annotated die photograph showing the four AI cores, CMU, PLL, and the clock generation unit. The AI core was implemented using a modular design approach with multi-level design hierarchy partitioned to carefully balance implementation complexity and design quality with overall turnaround time, while care was taken with the core floorplan to minimize wire length and unnecessary fan-out between partitions. To facilitate chip integration, the ring bus was designed to connect the cores via abutment. Tightly coupled logic and physical design optimizations enabled the core to achieve low-voltage operation while balancing product-driven throughput and energy-efficiency requirements. As shown by the measured shmoo plot in Fig. 14(b), the chip demonstrates

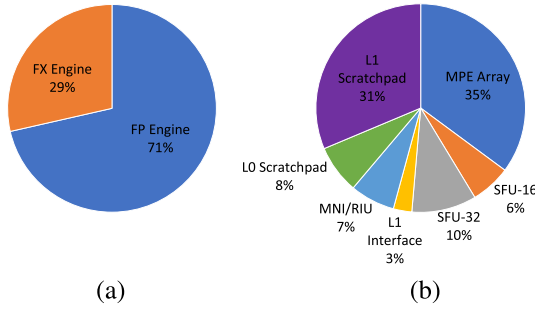


Fig. 15. Area breakdown of (a) MPE and (b) core.

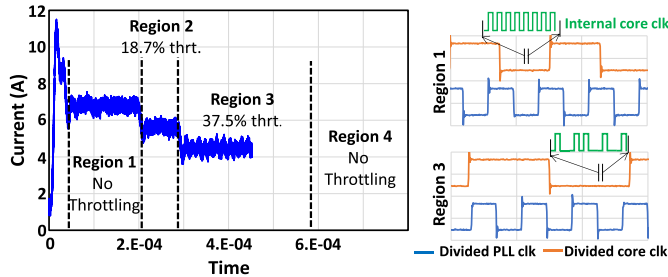


Fig. 16. Measured oscilloscope waveform of chip current consumption with varying throttling rates and measured core clock frequency (divided-by-1024) demonstrating reduction in average frequency due to throttling.

1.0-GHz operation at 0.55-V core (V_{DD}) and 0.7-V SRAM voltage (V_{SRAM}), scaling up to 1.6-GHz operation at 0.75-V V_{DD} and 0.95-V V_{SRAM} . Fig. 15 presents the area breakdown of the MPE and AI core. The core area shows the design balance between differing compute requirements (convolution and matrix multiplication versus vector operations) and local storage requirements.

V. MEASUREMENT RESULTS

A. Clock Frequency Throttling via Clock-Edge Skipping

Fig. 16 plots the chip current and clock measurements to demonstrate the operation of the clock frequency throttling implemented on the AI chip. The oscilloscope current waveform shows the change in chip current as the throttle rate is cycled from 0% to 18%, then 37%, and back to 0%, demonstrating the expected reduction in current draw of the chip based on the throttle rate. The oscilloscope trace is also presented, showing the divided-by-1024 core clock and PLL clock, which shows the reduction in divided clock frequency when throttling is turned on to demonstrate the average frequency reduction effects.

B. Performance Measurements

To measure the chip performance, the chip was tested using a comprehensive library of DL kernels supported by our software stack [25]. These lightweight kernels provide an agile way of benchmarking the chip performance while providing full coverage of all operations necessary to combine to run end-to-end AI workloads. Fig. 17 presents the power breakdown between compute and SRAM access in the cores

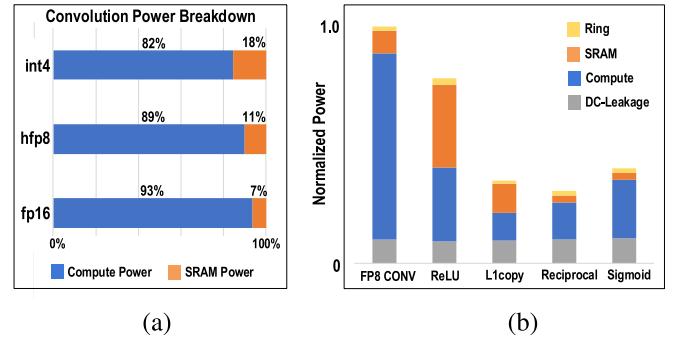


Fig. 17. (a) Comparison of measured power breakdown between compute and SRAM access for FP16, FP8, and INT4 formats running convolution. (b) Measured power breakdown of different components on the chip running various kernels on four cores.

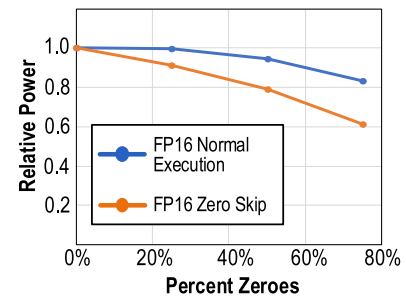
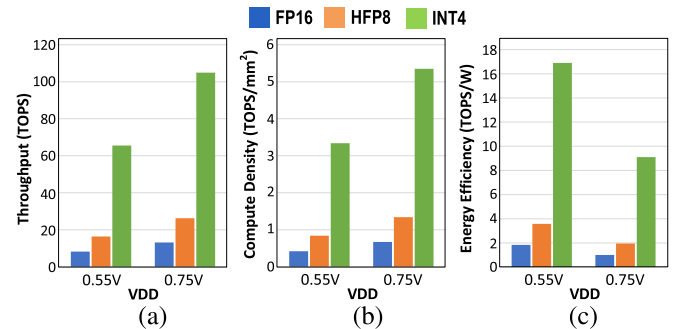


Fig. 18. Measured power savings from zero-skipping logic.

Fig. 19. Comparison of measured (a) throughput (TOPS), (b) compute density (TOPS/mm²), and (c) energy efficiency (TOPS/W) for FP16, HFP8, and INT4 formats spanning the core operating range.

for FP16, HFP8, and INT4 convolution, and the power breakdown of the different components for various commonly used kernels, including convolution, activation functions (ReLU and sigmoid), reciprocal, and data copy to L1 (L1Copy). As shown in Fig. 17(a), SRAM access power increases relative to compute at lower precisions, as higher throughput requires more frequent SRAM accesses. Fig. 18 plots the measured power savings from the zero-skipping logic that bypasses the FPU pipeline when one of the multiplicands is zero, showing 20% power savings for FP16 computation when the activation sparsity is 50%.

Measured throughput, compute density, and energy efficiency summarizing the peak performance are presented in Fig. 19 for the two voltages spanning the measured

TABLE I
COMPARISON WITH RELATED WORK

	This work*	Oh, IBM [23]	Lee, KAIST [5]*	Park, SNU [27]	Jiao, Alibaba [7]	Lin, MediaTek [28]*	A100, Nvidia [29]*
Technology	7nm	14nm	65nm	40nm	12nm	7nm	7nm
Purpose	CNN, FC, RNN	CNN, FC, RNN	CNN, FC, RNN	CNN, FC, RNN	CNN, FC	CNN, FC	GPU
Supply Voltage (V)	0.55 0.75	0.54 0.62	0.78 1.1	0.75 1.1		0.575 0.825	
Frequency (GHz)	1.0 1.6	1.0 1.5	0.05 0.2	0.02 0.18	0.7	0.29 0.88	1.14 (boost)
Area (mm ²)	19.6	9.8	16	6.25	709	3.04	826
On-chip SRAM (MB)	8	2	0.372	0.293	192	2.125	40 (L2)
Available Precision	Hybrid-FP8, FP16, FP32 INT4, INT2	FP16, FP32	FP8, FP16**	FP8-SEB	INT8, INT16 FP8, FP16, FP24	FP16, BF16 TF32, FP32, FP64, INT8, INT4	ASYMM-Q8, INT8 INT16, FP16
Floating Point Perf. (TFLOPS)	16.38 26.2 ¹	2.0 3.0 ²	0.15 0.6 ¹	0.063 0.567 ¹			312 ²
TFLOPS/mm ² (FP)	0.84 1.34 ¹	0.20 0.30 ²	0.0094 0.038 ¹	0.01 0.91 ¹			0.38 ²
TFLOPS/W (FP)	3.58 1.95 ¹	1.4 1.1 ²	3.5 1.64 ¹	4.81 2.5 ¹			0.78 ²
Integer Perf. (TOPS)	65.5 104.9 ⁴				825 ³	1.19 3.6 ³	1248 ⁴
TOPS/mm ² (INT)	3.35 5.35 ⁴				1.16 ³	0.39 1.19 ³	1.5 ⁴
TOPS/W (INT)	16.9 9.1 ⁴				2.99 ³	6.83 3.42 ³	3.12 ⁴
Scalability Mechanism	Memory Neighbor Interface, Bi-directional Ring	Memory Neighbor Interface	Single Chip	Single Chip	Uni-directional Ring, PCIe	Single Chip	Chip with NVLink or PCIe

* Reported performance at 0% sparsity

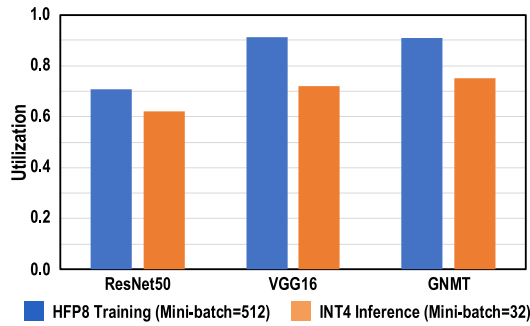
** Fine-grained mixed precision format [5]
Precision: ¹FP8 ²FP16 ³INT8 ⁴INT4

Fig. 20. Projected sustained utilization for a scaled-up 32-core AI chip.

range. With each MPE performing 32 HFP8 operations (16 MULs + 16 ADDs) or 128 INT4 operations (64 MULs + 64 ADDs) per cycle, the chip achieves the peak throughput of 26.2-TFLOPS and 104.9-TOPS for HFP8 and INT4 modes, respectively, operating at 1.6 GHz at 0.75-V V_{DD} . The chip achieves the peak energy efficiency of 3.58 TFLOPS/W in the HFP8 mode and 16.9 TOPS/W in the INT4 mode operating at 1 GHz at 0.55-V V_{DD} . This chip achieves leading-edge performance in energy efficiency and compute density across FP16, FP8, and INT4 modes and, compared to previously published works (Table I), achieves a product-driven balance of throughput and efficiency while supporting multiple precisions for low-precision training and inference without model accuracy degradation.

C. End-to-End Sustained Utilization

While peak performance is an important metric for hardware accelerators, it is equally important for end-to-end application performance to maintain high sustained utilization of the peak performance across a wide spectrum of application parameters, especially when scaling to large-scale many-core systems.

For that purpose, we expand this work to study a projected scaled-up system with 32-cores as described in Section III-H. Fig. 20 presents the projected end-to-end sustained utilization for training and inference for ResNet-50, VGG16, and GNMT, demonstrating >70% utilization for ResNet-50 and over 90% utilization for VGG16 and GNMT for training in the HFP8 mode, while INT4 inference shows 62% utilization for ResNet-50 and >70% utilization for VGG16 and GNMT.

VI. CONCLUSION

This article describes a 7-nm four-core mixed-precision AI chip that demonstrates leading-edge power efficiency for low-precision training and inference without model accuracy degradation. The chip leverages cutting-edge algorithmic advances and aggressive architecture/circuit optimizations to implement iso-accurate HFP8 training and INT4 inference to achieve 3.58 TFLOPS/W for HFP8 training and 16.9 TOPS/W for INT4 inference, respectively, while maintaining high compute utilization. In addition, high-bandwidth ring interconnect enables efficient data transfers for many-core scalability, while workload-aware power management with clock frequency throttling maximizes the application performance within a given power envelope.

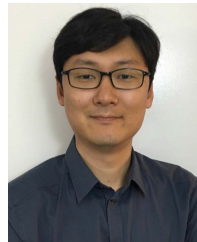
ACKNOWLEDGMENT

The authors would like to thank C. Baks, J. Jones, R. Oldrey, P. Gaschke, and D. Turnbull for design and characterization support and J. Burns, T.-C. Chen, and M. Khare for executive support.

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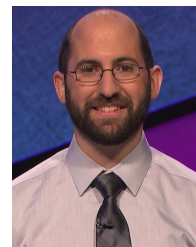
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She, an IBM Distinguished Engineer, is leading the development of competitive arithmetic units for POWER and z Systems. She has been driving a shift in IBM's systems value toward stack solutions based on hardware/software co-design and co-optimizations, specialized arithmetic accelerator engines, and hardware differentiation. Prior to joining

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Martin Lutz, photograph and biography not available at the time of publication.



Jinwook Jung (Member, IEEE) received the B.E. and M.E. degrees in computer engineering from Kobe University, Kobe, Japan, in 2011 and 2013, respectively, and the Ph.D. degree in electrical engineering from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 2018.

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Dr. Jung has served on the Technical Program Committee for the International Conference On Computer-Aided Design (ICCAD), the Asia and South Pacific Design Automation Conference (ASP-DAC), and the International Symposium on Physical Design (ISPD). He has been serving as the Chair for the IEEE Council on Electronic Design Automation (CEDA) Design Automation Technical Committee since 2021.



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He has been with the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, since 2008, where he currently works on the design and verification of artificial intelligence (AI) hardware accelerators for deep learning applications.



Ching Zhou (Member, IEEE) received the Ph.D. degree in physics from Université Paris-Sud, Orsay, France, in 1994.

He joined IBM in 1995 first as a post-doctor at the Physics Department working on quantum transistors and then in 1997 as an RSM at the VLSI Department. He is currently a Research Staff Member (RSM) at the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA. He has worked on many generations of P/Z/Bluegene microprocessor and deep learning accelerator design projects.

He initiated converting handcrafted custom adders to synthesizable designs in IBM servers, trailblazed and implemented first synthesized functional units floating point unit (FPU) and decimal floating point unit (DFU), and first neuron-network-based circuitry for IBM's high-performance systems. He has four issued patents and a few dozens of peer-reviewed publications in quantum physics, microprocessor and accelerator designs, and related methodologies.



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on process design kit (PDK) library development and support for advanced process nodes, including nanosheets and vertical FETs, as well as phase change memory (PCM) devices for IBM analog artificial intelligence (AI) cores. She is currently working on AI accelerators.



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Robert Casatuta, photograph and biography not available at the time of publication.



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Jungwook Choi received the B.S. and M.S. degrees in electrical and computer engineering from Seoul National University, Seoul, South Korea, in 2008 and 2010, respectively, and the Ph.D. degree in electrical and computer engineering from the University of Illinois at Urbana-Champaign, Champaign, IL, USA, in 2015.

He worked at the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, as a Research Staff Member from 2015 to 2019. He is currently an Assistant Professor at Hanyang University, Seoul. His main research interest is efficient implementation of deep learning algorithms.

Dr. Choi has received several research awards, such as the Design Automation Conference (DAC) 2018 Best Paper Award, and has actively contributed to the academic activities, such as the Technical Program Committee Co-Chair/Chair of the Design, Automation and Test in Europe Conference (DATE) 2021 and DAC 2021.



Kyu-Hyoun (KH) Kim (Member, IEEE) received the Ph.D. degree in electrical engineering from the Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea, in 1997.

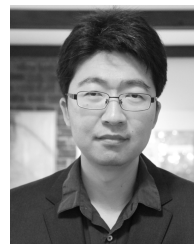
In 1998, he joined Samsung Electronics, Hwaseong, South Korea, and led the I/O Circuit Design Team for DDR1/DDR2/DDR3 SDRAM and Graphics Memories. He joined the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, in 2006, and led memory subsystem development for IBM HPC Systems, including BlueGene and Exascale systems. He is currently in charge of exploratory artificial intelligence (AI) hardware. He represents IBM in JEDEC for memory standardization. He holds 176 issued U.S. patents.

Dr. Kim has been a member of the ISSCC International Technical Program Committee (ITPC) since 2019. He has presented eight papers to the International Solid-State Circuits Conference (ISSCC) as first authors. He received the ISSCC Takuo Sugano Outstanding Paper Award in 2007, the JEDEC Technical Recognition Award in 2011, and the JEDEC Award of Excellence in 2019.



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He joined IBM, East Fishkill, NY, USA, in 2001. He has been working at the IBM Semiconductor Research Development Center (SRDC), East Fishkill, and the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA. His research interests are centered on device physics, process integration, and electrical characterization of MOSFETs. His experience includes device simulation, novel device design and integration, carrier mobility characterization, silicon-on-insulator (SOI) CMOS scaling, technology development, and production implementation. He recently started working on chip design.



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He then joined IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, studying electrical and optical properties of superconducting films. In 1985, he joined the VLSI Test and Measurement Group, where he worked on picosecond opto-electronics and developed a high-bandwidth probe for testing high pin count integrated circuits. In 1995, he joined the VLSI Design Group, where he was responsible for

the L2 integration on the POWER4 and POWER5 microprocessors. He then served as the lead integrator for the POWER6 core. Subsequently, he worked on the high-level design for the Series P and BlueGeneQ microprocessors and 3-D integration for high-performance systems. Recently, he has been working on integration and power analysis for next-generation artificial intelligence (AI) computer chips. He has coauthored more than 49 technical articles. He holds 11 U.S. patents.

Dr. Scheuermann received various awards at IBM, including the Research Division Award, eight Outstanding Technical Achievement Awards, and four Invention Achievement Awards.



Xiao Sun received the B.S. degree from Peking University, Beijing, China, in 2008, and the M.S. and Ph.D. degrees from Yale University, New Haven, CT, USA, in 2010 and 2013, respectively, advised by Prof. T. P. Ma.

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Dr. Sun was awarded “35 innovators under 35” by MIT Tech Review in 2021.



Hung Tran received the B.Tech. and M.Eng. degrees from the Institute of Technology Hanoi, Hanoi, Vietnam, in 1996 and 1998, respectively, and the Dr.Eng. degree from the Dresden University of Technology, Dresden, Germany, in 2006.

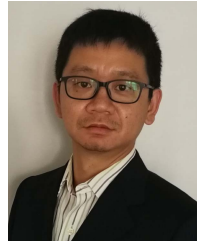
In 2007, He joined IBM Microelectronics, East Fishkill, NY, USA, working on various technologies from BiCMOS8HP to 5-nm CMOS. Since 2014, he has been with the IBM T. J. Watson Research Center, Yorktown Heights, NY, USA, working on exploratory semiconductor research. Since 2019,

he has been working on artificial intelligence (AI) accelerator research. He holds 27 U.S. and international patents.



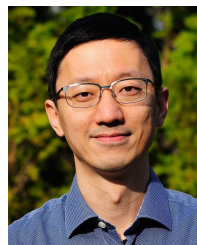
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Dr. Zhang has served as a Guest Editor for IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS (JETCAS) and IEEE SOLID-STATE CIRCUITS LETTERS (SSC-L). He is currently serving as a Technical Program Committee Member for the Applied Power Electronics Conference (APEC), the IEEE VLSI Symposium on Technology and Circuits, the IEEE Custom Integrated Circuits Conference (CICC), and the IEEE International Solid-State Circuits Conference (ISSCC). He is also serving as a Technical Advisory Board Member for the Analog-Mixed Signal Circuits, Systems, and Devices (AMS-CSD), Semiconductor Research Corporation (SRC).



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After joining the IBM Systems Group, Hursley, U.K., in 1984, he initially worked at the Circuit Design Group and then moved to the IBM Storage group, involved in the design of the first CMOS chips developed by IBM Storage and then worked

on the development of high-speed serial interfaces, including the definition and implementation of IBM's serial storage architecture. He then took on application-specific integrated circuit (ASIC) team lead roles and also tracked chip verification activities and eventually transitioned to the ASIC project management. He was the project manager for storage controller ASICs developed for use in IBM's DS8000 product and the project manager for the memory controller used in IBM Research's BlueGeneQ ASIC. Subsequently, he was a project manager for a couple of RAID controller ASICs developed by the IBM's Server ASIC Team for use on adapter cards shipped with Power Systems and DS8000. He was a project manager involved in the development of the memory buffer ASIC, which is deployed in the Power 10 Server and the latest zSeries Servers. He is currently a Project Manager with IBM Systems ASICs.

Brian Curran, photograph and biography not available at the time of publication.



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Pong-Fei Lu received the B.S.E.E. degree from National Taiwan University, Taipei, Taiwan, in 1978, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 1986.

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Dr. Lu received two Outstanding Technical Awards for his work in the design of S390 and Power5 microprocessors.



Sunil Shukla received the Ph.D. degree in electrical engineering from The University of Queensland, Brisbane, QLD, Australia, in 2008.

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Kailash Gopalakrishnan received the Ph.D. degree in electrical engineering from Stanford University, Stanford, CA, USA, in 2004.

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Leland Chang (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electrical engineering and computer sciences from the University of California at Berkeley, Berkeley, CA, USA, in 1999, 2001, and 2003, respectively.

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Dr. Chang is the former Memory Subcommittee Chair of the ISSCC Technical Program Committee.